

5.2 What factors are important to consider when selecting appropriate materials and/or components when designing?

- a. The characteristic properties of the **main categories** of materials (5.1 a–e, not the specific materials) and why this makes them appropriate for different uses, such as:
- density, strength, hardness, durability, strength to weight ratio, stiffness, elasticity, impact resistance, plasticity, corrosive resistance to chemicals and weather, flammability, absorbency, thermal and electrical conductivity.



	b. The physical and working properties of specific materials and/or system components, with consideration of: i. how easy they are to work with ii. how well they fulfil the required functions of products in different contexts.	
	c. Other factors that influence the selection of materials and/or components, including; i. required functionality of the design solution ii. aesthetic attributes iii. environmental considerations iv. availability and cost of stock forms v. social, cultural and ethical considerations	